

Question Number	Answer	Additional Guidance	Mark
1a	An answer that includes the following points. • {α / alpha} glucose molecules (1) • (which are) joined by glycosidic bonds (1) • 1,4 bonds (1) • amylose AND amylopectin (1) • {amylose, coiled / amylopectin, branched or have 1,6 bonds} (1)	accept anywhere EXCEPT in context of branched accept mixture of	3

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1bi	An explanation that includes two of the following points. • more starch is synthesised above 20 / 25°C and below 30°C (than at any other temperatures) (1) • (because the manufacture of starch) will involve enzymes (1) • explanation of the effect of temperature change on enzyme activity (1)	accept most made between 20 / 25 and 30 accept if implied e.g. effect of temperature on number of {successful collisions of E and S / number of ES complexes effect of temperatures over {30 /	2

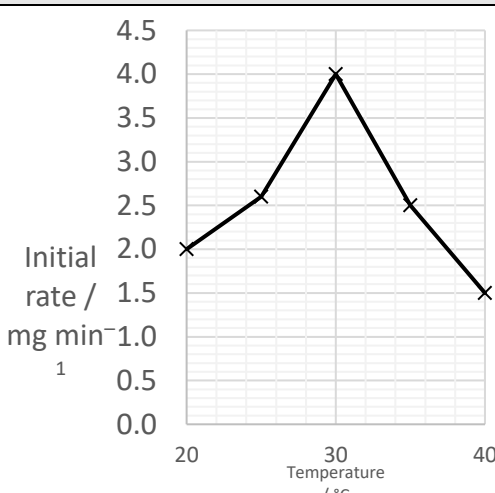
		optimum} on enzyme shape / ref denaturation}	
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1bii	A table with the following features: • suitable table drawn (1) • headings with units (1) • only means entered correctly (1)	<table><tr><th>Temperature / °C</th><th>Mean rate of starch synthesis / mg day⁻¹</th></tr><tr><td>20</td><td>6.1</td></tr><tr><td>25</td><td>7.3</td></tr><tr><td>30</td><td>7.6</td></tr><tr><td>35</td><td>6.4</td></tr></table> Mp 2 lost if units in cells Mp3 lost if SDs included	Temperature / °C	Mean rate of starch synthesis / mg day ⁻¹	20	6.1	25	7.3	30	7.6	35	6.4			3
Temperature / °C	Mean rate of starch synthesis / mg day ⁻¹														
20	6.1														
25	7.3														
30	7.6														
35	6.4														

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1biii	An answer that includes: - value of SD: 0.35 (mg day⁻¹) (1) plus Explanation, one of : - when applied to the mean it {overlaps with value at 25 (°C) / does not overlap with value at 35 (°C)} (1) (a measure of) the {variation / dispersion / spread / deviation / fluctuation / variability } (of data) around the mean (1)	Ignore any sign a region that includes 68 % of all the data points there is a 32 % chance of the value being higher or lower by 0.35 do not accept range, margin of error	2

Question Number	Answer	Additional Guidance	Mark
1biv	An answer that includes the following points: • difference between 20 (°C) and 25 (°C) is significant because SDs do not overlap (1) • difference between 25 (°C) and 30 (°C) is not significant because SDs do overlap (1)	accept error bars for SDs for all mps do not accept range bars	2

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1ci	4.0 (mg min ⁻¹) (1)	accept 4	1

Question Number	Answer	Additional Guidance	Mark
1cii	A graph with the following features • A axes correct (x - temperature, y- initial rate) (1) • L all labels correct (x- temperature / °C, y / initial rate mg min⁻¹ / mg/min) (1) • P plots correct on a linear scale on both axes (1) • S points joined with straight lines (1)	 Ignore plot at 30 °C lose mp P If extend to 0 or 10 C lose mp S extrapolate beyond 40 does not have to start at zero on either axis but must what it does start at bar charts lose mp S	4

Question Number	Answer	Additional Guidance	Mark
1ciii	An answer that includes any two of the following points: • (the initial rate) would not be influenced by a {change / reduction} in substrate concentration (1) • (the initial rate) would not be influenced by a {change / build-up} of products (1)	accept substrate concentration would not be a limiting factor accept substrate used up	2

	• allows (valid) comparisons of rates (between temperatures) (1)		
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